Definition 12.1

Any operation $*$ defined on a non-empty set S is called a **binary operation** on S if the following conditions are satisfied:

(i) The operation $*$ must be defined for each and every ordered pair $(a,b) \in S \times S$.

(ii) It assigns a **unique** element $a*b$ of S to every ordered pair $(a,b) \in S \times S$.

In other words, any binary operation $*$ on S is a rule that assigns to **each ordered pair** of elements of S a **unique** element of S. Also $*$ can be regarded as a **function (mapping)** with input in the Cartesian product $S \times S$ and the output in S.

$$*: S \times S \to S \ ; \ *(a,b) = a*b \in S$$, where $a*b$ is an unique element.

A binary operation defined by $*: S \times S \to S$; $*(a,b) = a*b \in S$ demands that the output $a*b$ must always lie the given set S and not in the complement of it. Then we say that '$*$ is closed on S' or 'S is **closed** with respect to $*$'. This property is known as the **closure property.**

Definition 12.2

Any non-empty set on which one or more binary operations are defined is called an **algebraic structure.**

Another way of defining a binary operation $*$ on S is as follows: $\forall a,b \in S$, $a*b$ is unique and $a*b \in S$.

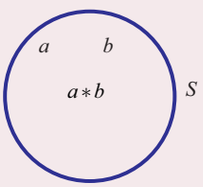


Fig. 12.1

Note

It follows that every binary operation satisfies the closure property.

Note

The operation $*$ is just a symbol which may be $+, \times, -, \div$ matrix addition, matrix multiplication, etc. depending on the set on which it is defined.

For instance, though $+$ and $\times$ are binary on $\mathbb{N}$, $-$ is **not** binary operation on $\mathbb{N}$.

To verify this, consider $(3,4) \in \mathbb{N} \times \mathbb{N}$.

$$*(a,b) = -(3,4) = 3-4 = -1 \notin \mathbb{N}.$$

Hence $-$ is **not binary operation** on $\mathbb{N}$. So $\mathbb{N}$ is to be extended to $\mathbb{Z}$ in order that $-$ becomes binary operation on $\mathbb{Z}$. Thus $\mathbb{Z}$ is closed with respect to $+, \times,$ and $-$. Thus $(\mathbb{Z}, +, \times, -)$ is an algebraic structure.

Observations

The binary operation depends on the set on which it is defined.

(a) The operation – which is **not binary operation** on $\mathbb{N}$ but it is binary on $\mathbb{Z}$. The set $\mathbb{N}$ is extended to include negative numbers. We call the included set $\mathbb{Z}$.

(b) The operation $\div$ on $\mathbb{Z}$ is **not binary operation** on $\mathbb{Z}$. For instance, for $(1,2) \in \mathbb{Z} \times \mathbb{Z}$, $\div(1, 2) = \frac{1}{2} \notin \mathbb{Z}$. Hence $\mathbb{Z}$ has to be extended further into $\mathbb{Q}$.

(c) It is a known fact that the division by 0 is **not** defined in basic arithmetic. So $\div$ is binary operation on the set $\mathbb{Q} \setminus \{0\}$. Thus $+, \times, -$ are binary operation on $\mathbb{Q}$ and $\div$ is binary operation on $\mathbb{Q} \setminus \{0\}$.

Now the question is regarding the reasons for extending further $\mathbb{Q}$ to $\mathbb{R}$ and then from $\mathbb{R}$ to $\mathbb{C}$. Accordingly, a number system is needed where not only all the basic arithmetic operations $+, -, \times, \div$ but also to include the roots of the equations of the form "$x^2 - 2 = 0$" and "$x^2 + 1 = 0$".

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